Finite Uniform Input and Periodic Residual Costs for Smooth Isometric Immersions in Four-Space

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Abstract

The assigned manuscript claims smooth isometric immersions of all closed Riemannian surfaces in $\mathbb{R}^4$. Its exact correction uses finite oscillatory formulas followed by smoothing and an infinite succession of derivative-order blocks, not an ordinary bounded-inverse Newton theorem. We retain that mechanism and prove a conditional improvement of its finite uniform input: the six-error scheme can start at order 115 instead of 440. This number is minimal in a precisely specified common-order scale family, not an optimal differentiability theorem. We also identify the source's periodic covariance solve as the least-energy right inverse in its physical mean-zero L^2 space, obtaining exact residual costs and a degeneracy diagnostic. The primitive increment, later boundary inequalities, and global finite assembly are mapped separately from these deductions. No finite residual test is asserted to establish the source's global geometric conclusion. Contribution assessment: C3/5, a proved source-specific reduction of the finite uniform input order and exact periodic correction costs, conditional on the source's geometric and small-mode estimates.

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1 Geometric Problem and Proof Boundaries

Let M be a compact smooth two-manifold without boundary and let h_* be a smooth positive definite metric. Neither orientation nor connectedness is assumed. For a smooth immersion $F : M \to \mathbb{R}^4$, set

$$\gamma(F)_{ij} = \langle \partial_i F, \partial_j F \rangle , \qquad B_{ij} = \mathrm{proj}_{N_F} \partial_{ij} F, \quad N_F = (dF(TM))^{\perp}.$$

All C^m norms below use a fixed smooth background metric and a finite precompact atlas, with fixed norm equivalence constants. Normal norms and pairings are the actual Euclidean ones. The Gauss equation is

$$\langle B_{xx}, B_{yy} \rangle - |B_{xy}|^2 = K_{\gamma(F)} \det \gamma(F). \tag{1}$$

An immersion need not be injective. The source's global claim is the existence of F with $\gamma(F) = h_*$ for every such (M, h_*) [1]. This report does not turn that claim into an independently certified theorem.

The distinction matters already infinitesimally. The differential

$$(D\gamma_G Z)_{ij} = \langle \partial_i G, \partial_j Z \rangle + \langle \partial_j G, \partial_i Z \rangle$$

has infinitesimal rigid motions in its kernel. More importantly, the source's phase inverses lose spatial derivatives. An invertible finite moment matrix, an L^2 right inverse for a periodic auxiliary equation, or a small sampled metric residual is not a bounded inverse on the full smooth metric problem.

1.1 Source Architecture

The complete assigned text has been read, including its included modules. Its logical chain is as follows; the geometric inputs in this chain remain source inputs to the refinements proved later.

1. *Good phases and finite decompositions.* A covector dx is good when $B(Y, Y) \neq 0$ for $Y \in \ker dx \setminus \{0\}$. At a nonzero normal-valued quadratic form there are finitely many bad projective directions. Positive tensors are decomposed as $\sum b_\nu^2 d\phi_\nu^2$, with the needed sums, differences and doubles also good. Square roots are taken on a strictly positive tensor neighborhood before multiplication by cutoffs.

2. *Small modes.* A finite conjugated differential construction produces an approximate primitive metric increment. Both its finite derivative loss and its polynomial dependence on phase bounds are retained.

3. *Exact correction.* Signed moment smoothing controls all required derivatives of the smoothed inputs from a finite weighted norm. The finite increment is added to the true, unsmoothed map. Higher true derivatives then enter linearly. Infinitely many finite blocks raise the controlled order.

4. *Large primitive increment.* A smooth constant-length velocity loop gives the leading primitive metric exactly. Periodic correctors remove successive powers of the fast scale. A finite slow-map construction prepares the hypotheses of exact correction.

5. *Preservation and assembly.* Later boundary directions, both ordered crossing inequalities, and a preferred normal survive each primitive addition. A finite cycle decomposition, a prepared initial immersion, and generic boundary choices assemble the prescribed metric.

The actual metric is prescribed at every primitive endpoint. Conversely, the last exact correction is only globally C^2-small; it is not claimed to vanish outside the primitive support. This is why preservation of open geometric inequalities, rather than an unsupported exact support assertion, is essential.

1.2 Current Primary Context

Lewicka's current arXiv version is v3, August 14, 2026 [2]. Theorem 1.1, read directly, concerns a planar domain diffeomorphic to a disk, $g \in C^{r,\beta}$ with $0 < r+\beta \le 2$, and $C^{1,\alpha}$ approximation of short immersions for $\alpha < \min\{(r+\beta)/2, 1\}$. Appendix Theorem 9.1 states the smooth Poznyak theorem for a bounded simply connected planar domain and sketches its proof. Neither statement directly supplies the closed, possibly nonorientable global assembly in the assigned source. The v3 comment says that the normal-frame analysis is now self-contained, replacing reliance on a partly erroneous external result. This is version context, not our audit of that paper's complete proof. Classical Nash–Moser smoothing is the relevant conceptual setting [3]; no historical novelty is asserted for smoothing, Hilbert-space least-norm inversion, or power balancing.

2 What the Small-Mode Interface Actually Supplies

For $0 < s \le 1$, define

$$|f|_{m,s} = \sum_{j=0}^{m} s^j \, \|f\|_{C^j}\,, \qquad |j^2 G|_{m,s} = \sum_{j=0}^{m} s^j \, \|G\|_{C^{j+2}}\,.$$

The source's more general finite operator has the form

$$P(G) = \gamma(G) + \sum_{i=1}^{a} \varepsilon^i P_i(G),$$

where P_i is polynomial in higher jets with smooth low-jet coefficients. Its excess-derivative count gives a fixed exponent p, independent of the chosen finite accuracy q. With $\tau \le s$, δ/τ sufficiently small, and $\eta = \tau/s + \varepsilon\tau^{-p}$ sufficiently small, the source constructs a smooth U with

$$|U|_{m,\tau} \le C_m \delta\tau, \qquad |P(G+U) - P(G) - \delta^2 H|_{m,\tau} \le C_m\{\delta\eta^q + \delta^3/\tau\}. \tag{2}$$

Every fixed q, m uses only finitely many smoothed derivatives, but their order may grow with q, m. Section 4 uses only the unperturbed case $\varepsilon = 0$. It does not need a fixed finite derivative bound valid for all values of q simultaneously.

Here is the inverse-domain reason for that qualification. In a good phase x, write a complex amplitude as $Z = G_k g^{kl} u_l + v$, with v normal. Conjugating by $e^{ix/\tau}$ gives

$$(D^x_G Z)_{ij} = \partial_i u_j + \partial_j u_i - 2\Gamma^k_{ij} u_k + \frac{i}{\tau}(\delta_{ix} u_j + \delta_{jx} u_i) - 2\,\langle B_{ij}, v\rangle\,. \tag{3}$$

With $b = B_{yy} \neq 0$, the yy equation is solved by

$$v = v_0 + \frac{b}{|b|^2}(\partial_y u_y - \Gamma^k_{yy} u_k - h_{yy}/2), \qquad v_0 \perp b.$$

The remaining equations have a leading iu/τ term and differential lower terms. A finite residual iteration gains powers of τ/s while using more derivatives. Zero-phase nonlinear terms are corrected by finite substitution, and nonzero phases by the same finite differential formulas. An estimate of the form $\left\|T(A) - T(\bar{A})\right\|_m \le C\eta \left\|A - \bar{A}\right\|_{m+\ell_q}$ is not a Banach contraction in C^m. The source does not need to call it one.

2.1 Original Ram Comparison

The original stability chapter [4], read in full, proves radii-ball and perturbation conclusions under an actual bounded isomorphism between the declared domain and range, with derivative and residual bounds on a full ball. Those hypotheses are not silently supplied by (3). Its finite-rank residual tests also do not establish surjectivity of an infinite-dimensional differential operator. The freshly read original jet proofs [5] retain integral Taylor remainders, the complete Leibniz rule, and omitted derivative terms in finite recovery. They support the bookkeeping below: differentiating a small remainder requires bounds on its differentiated remainder, and recovering finitely many jets does not make them a globally compatible smooth field.

3 An Exact Periodic Energy and Residual Formula

This section isolates a genuine equation from the primitive construction. Let P be a two-dimensional Euclidean plane and let $V : S^1 \to P$ be a smooth loop with $|V(\theta)| = R > 0$. Averages and L^2 norms use normalized Haar measure $d\theta/(2\pi)$. Put

$$v = \operatorname{av} V, \quad Z = V - v, \quad \Sigma = \operatorname{av}(Z \otimes Z), \quad M = \mathrm{Id}_P - \Sigma/R^2, \quad \mu = \lambda_{\min}(M). \tag{4}$$

For scalar or vector functions let $\Pi f = f - \operatorname{av} f$. On the physical spaces $\mathcal{H} = L^2_0(S^1; P)$ and $\mathcal{K} = L^2_0(S^1; \mathbb{R})$ define

$$A : \mathcal{H} \to \mathcal{K}, \qquad AW = \Pi(V \cdot W).$$

The source solves $AW = r$, $\operatorname{av} W = 0$, for the mean-zero scalar remainder r appearing after the tangential part of a periodic corrector has been fixed.

Lemma 3.1. *For a continuous constant-length loop, M is positive definite. No positive lower bound for μ follows from R alone.*

Proof. For a unit $q \in P$,

$$R^2 \langle Mq, q\rangle = \operatorname{av}\{R^2 - (q \cdot V)^2\} + (q \cdot v)^2 \ge 0.$$

Equality would force $V \in \{Rq, -Rq\}$ everywhere by continuity. The circle is connected, so V is constant and $(q \cdot v)^2 = R^2$, a contradiction. Compactness of the unit circle gives $\mu > 0$. Smooth loops can spend almost all their time near Rq and $-Rq$, with nearly equal proportions and very short connecting arcs; their covariance tends to $R^2 q \otimes q$ and $\mu \to 0$. Thus pointwise invertibility is not a uniform gap theorem. □

Theorem 3.2 (Least physical energy)**.** *For $r \in \mathcal{K}$, let $b = \operatorname{av}(Vr)$ and $m = M^{-1}b$. The source's formula*

$$h = r + \frac{(V - v) \cdot m}{R^2}, \qquad W_* = (Vh - m)/R^2 \tag{5}$$

is the unique minimum-L^2-norm solution of $AW = r$. Moreover,

$$\|W_*\|_2^2 = \frac{\|r\|_2^2}{R^2} + \frac{b \cdot M^{-1} b}{R^4}, \tag{6}$$

$$\|W_*\|_2 \le \frac{\|r\|_2}{R\sqrt{\mu}}. \tag{7}$$

Every other solution is $W_* + W_0$, $W_0 \in \ker A$, with squared norm $\|W_*\|_2^2 + \|W_0\|_2^2$. If an approximate mean-zero $\widehat{W}$ has residual $e = r - A\widehat{W}$, the least-energy additive correction is the same formula applied to e, with the exact cost (6).

Proof. The adjoint in these spaces is $A^* r = Vr - \mathrm{av}(Vr)$. If $U : P \to \mathcal{K}$ is $Uq = Z \cdot q$, then $U^* r = \mathrm{av}(Vr)$ and $U^* U = \Sigma$. Direct calculation gives

$$AA^* = R^2 \mathrm{Id}_{\mathcal{K}} - UU^*, \qquad (AA^*)^{-1} = R^{-2}\mathrm{Id}_{\mathcal{K}} + R^{-4} U M^{-1} U^*.$$

Multiplication verifies the inverse, since $M = \mathrm{Id} - \Sigma/R^2$; all operators are bounded and $AA^* \ge R^2 \mu \mathrm{Id}$. Thus $A^*(AA^*)^{-1} r$ is a solution orthogonal to $\ker A$. Expanding it and using $b + \Sigma m / R^2 = m$ yields (5). Taking its energy as $\langle r, (AA^*)^{-1} r \rangle$ proves (6); the lower spectral bound proves (7). Orthogonal decomposition proves the rest. □

This is an interpretation and quantitative refinement of the source's already correct inverse, not a new existence mechanism. For a compact smooth parameter family of these loops, positivity gives a uniform gap only after the family is fixed. Spatial differentiation incurs, for example, $\partial M^{-1} = -M^{-1}(\partial M) M^{-1}$ and higher products; the L^2 bound alone gives no uniform control of those derivatives. If a certified covariance enclosure gives $\left\| M - \widehat{M} \right\| \le d$ and $\lambda_{\min}(\widehat{M}) > d$, the usable gap is $\lambda_{\min}(\widehat{M}) - d$. Computing a finite quadrature matrix without a bound on the missing integral is not such an enclosure.

Example 3.3 (Sharp cost and a negative control)**.** Use four unit vectors $(1,0), (-1,0), (0,1), (0,-1)$ with weights $(1-\epsilon)/2, (1-\epsilon)/2, \epsilon/2, \epsilon/2$, where $0 < \epsilon \le 1/2$. This is a finite probability analogue of the same physical spaces, not a substituted smooth velocity loop. Here $v = 0$, $M = \mathrm{diag}(\epsilon, 1-\epsilon)$. For $r = (1, -1, 0, 0)$,

$$\|r\|_2^2 = 1 - \epsilon, \qquad \|W_*\|_2^2 = (1-\epsilon)/\epsilon.$$

The factor $1/\sqrt{\mu}$ is attained. At $\epsilon = 0$ the two-point law has $A^* r = 0$ for this nonzero r, so the equation is not onto. Constant loops, in contrast, have $\Sigma = 0$, $M = \mathrm{Id}$, and no singularity. Vanishing primitive amplitude must not be confused with the two-point degeneracy.

4 Reducing the Finite Uniform Input Order

We now retain the source's nonlinear correction architecture, including its infinite order-raising schedule. All statements in this section are conditional on the source's good-phase construction and the small-increment estimate (2), with their stated polynomial bounds. The arithmetic, smoothing transfer, initialization and scheduling below are proved here. This is a refinement of the source's exact-correction proposition, not a replacement proof of all its geometric prerequisites.

4.1 Smoothing and the True-Map Identity

A signed compact smoothing kernel with moments $1, \dots, r$ equal to zero is obtained from finitely many dilations of a fixed kernel by a Vandermonde system. Positivity of the kernel is not asserted or needed. In a fixed atlas, with the usual cutoff extensions, the source proves

$$\|S_s u\|_{C^{q+m}} \le C s^{-m} \|u\|_{C^q}, \qquad \|u - S_s u\|_{C^{q+j}} \le C s^{r-j} \|u\|_{C^{q+r}} \quad (0 \le j \le r).$$

Define

$$\mathcal{S}_m(t; G, H) = \sum_{j=0}^{m} t^j (\|G\|_{C^{j+2}} + \|H\|_{C^j}).$$

If $\mathcal{S}_r \le B$ and $\tau \le s \le t$, splitting derivatives below and above r gives, for each fixed m,

$$|j^2 G_s|_{m,s} + |H_s|_{m,s} \le C_{m,r} B, \tag{8}$$

$$|j^2 (G - G_s)|_{m,\tau} + |H - H_s|_{m,\tau} \le C_{m,r} \{B(s/t)^r + \mathcal{S}_m (\tau/t)^r\}. \tag{9}$$

For derivatives up to r, the moment remainder supplies s^{r-j} and $\tau^j \le s^j$; for higher derivatives the true and smoothed inputs are bounded separately and $\tau^j / t^j \le (\tau/t)^r$. This proves the relevant two parts of (9). Estimate (8) follows by smoothing above r and ordinary boundedness below it. Constants can depend on m, r, which is harmless at each fixed block.

Set $H = \delta^{-2}(h_* - \gamma(G))$ and construct U at G_s with target $H_s - (\delta'/\delta)^2 h_*$. Let

$$E = \gamma(G_s + U) - \gamma(G_s) - \delta^2 [H_s - (\delta'/\delta)^2 h_*].$$

Add this U to G, not to G_s, and set $H' = \delta'^{-2}(h_* - \gamma(G + U))$. For $\mathcal{C}(V, W)_{ij} = \langle \partial_i V, \partial_j W \rangle + \langle \partial_j V, \partial_i W \rangle$, quadratic expansion yields exactly

$$\delta'^2 (H' - h_*) = \delta^2 (H - H_s) - E - \mathcal{C}(G - G_s, U). \tag{10}$$

There is no $(d(G - G_s))^2$ term. It cancels because only metric *increments* are compared. Thus every high true-input norm in this identity remains linear, which is the central reason the later waiting argument works. Replacing G by $G_s + U$ would change this bookkeeping and would require a separate displacement analysis.

Proposition 4.1 (General scale ledger)**.** *Take $\beta > 1$, $1 < \lambda < \beta$, $\tau = t^\beta$, $s = t^\lambda$, $\delta = t^k$, and $\delta' \in \{\tau^k, \tau^{k+1}\}$. Use moment order r and finite-mode accuracy q. After division by the more demanding target $\tau^{2(k+1)}$, the six error exponents are*

term	*exponent of t*
$\delta(\tau/s)^q$	$k + (\beta - \lambda)q - 2\beta(k+1)$
δ^3/τ	$(3 - 2\beta)k - 3\beta$
$\delta^2 B(s/t)^r$	$2k + (\lambda - 1)r - 2\beta(k+1)$
$\delta^2 \mathcal{S}_m(\tau/t)^r$	$2k + (\beta - 1)r - 2\beta(k+1)$
$\delta B(s/t)^r$	$k + (\lambda - 1)r - 2\beta(k+1)$
$\delta \mathcal{S}_m(\tau/t)^r$	$k + (\beta - 1)r - 2\beta(k+1)$.

(11)

If all these exponents are at least $\alpha = \beta - 1 > 0$, and $k - \beta \ge \alpha$, then the source step has the revised bound

$$\mathcal{S}_m(\tau; G + U, H') \le \|G\|_{C^2} + \|h_*\|_{C^0} + C_m(z, k) t^\alpha (1 + \mathcal{S}_m(t; G, H)). \tag{12}$$

Its constants and threshold remain independent of higher true-input norms, and polynomial in z^{-1} at fixed k, m under the source's phase bounds.

Proof. The first two rows are (2) with $\varepsilon = 0$. The next two use (9) on $H - H_s$, multiplied by δ^2. The last two use it on $G - G_s$ and $|dU|_{m,\tau} \le C_m\delta$. This proves the table and the corresponding defect bound from (10). Inherited positive-order weighted derivatives of G gain at least $\tau/t = t^{\beta-1}$; its new derivatives satisfy $|j^2U|_{m,\tau} \le C_m\delta/\tau$. The fixed tensor contributes $\|h_*\|_{C^0} + O_m(\tau)$. These observations prove (12). All derivatives needed by the finite-mode formulas are supplied by (8), whatever their finite order. The moment order r is not required to dominate that order; smoothing is doing that work. Polynomial constants and strictly positive powers give the claimed thresholds. □

Theorem 4.2 (Finite uniform input 115)**.** *In the source's exact-correction interface, replace the initial controlled order $R = 440$ by $R = 115$. More explicitly, fix h_* and retain the source's compact metric bounds, quantitative nonzero-second-form condition, and polynomial good-phase and small-mode estimates in a C^2 ball of radius $\rho_z = z^4$ about f_z. Assume each f_z is smooth, and uniformly as $z \downarrow 0$,*

$$\|f_z\|_{C^j} = O(z^{-j}) \quad (0 \le j \le 117), \qquad \|\gamma(f_z) - h_*\|_{C^{115}} = O(z^N). \tag{13}$$

There is an integer N, chosen sufficiently large from the polynomial threshold exponents, for which the same construction produces smooth $\widetilde{f}_z$ satisfying $\gamma(\widetilde{f}_z) = h_*$ and $\left\|\widetilde{f}_z - f_z\right\|_{C^2} = O(z^4)$. No uniform bound on the derivatives of f_z above order 117 is required. They must nevertheless exist.

Proof. Use

$$\beta = 11/9, \quad \lambda = 10/9, \quad k \ge 7, \quad r(k) = q(k) = 13k + 24, \quad \alpha = 2/9. \tag{14}$$

The six rows of (11), in order, become

$$\frac{2}{9}, \quad \frac{5k-33}{9}, \quad k + \frac{2}{9}, \quad \frac{22k+26}{9}, \quad \frac{2}{9}, \quad \frac{13k+26}{9}. \tag{15}$$

They are at least $2/9$ for $k \ge 7$. Also $k - \beta \ge 2/9$, $\delta/\tau \to 0$, $\tau/s \to 0$, and $\delta'/\delta \to 0$ for both choices of δ'. The last property keeps the smoothed target tensor in its fixed positive neighborhood. The same phase ball and smoothing estimates therefore apply. There is no use here of the numerical lower bound 10 except where the source had checked these inequalities with its old scales.

Put $B = z^{-4}$. The admissible C^2 ball gives $\|G\|_{C^2} + \|h_*\|_{C^0} < B/4$ for small z. For each fixed k, choose the scale threshold to preserve $\mathcal{S}_{r(k)} \le B$, the defect neighborhood, and the phase ball. Impose as well

$$\|U\|_{C^{\lfloor k/2 \rfloor}} \le \rho_z t. \tag{16}$$

This is possible because (2) gives the exponent $k + \frac{11}{9}(1 - \lfloor k/2 \rfloor) > 1$ for every $k \ge 7$. The thresholds cover both choices of δ' and are polynomial at the initial fixed block.

Choose $\mu_0 > 1$ large enough for all initial thresholds at $t = z^{\mu_0}$, then choose an integer $N > 14\mu_0 + 1$. Start with $k = 7$, $\delta = t^7$ and

$$G = \sqrt{1 - \delta^2}\, f_z, \qquad H = h_* + (1 - \delta^{-2})(\gamma(f_z) - h_*).$$

Thus $H \to h_*$ in C^{115}. Since $r(7) = 115$, the required weighted map derivatives are bounded by $Cz^{-2}\sum_{j=0}^{115} z^{(\mu_0-1)j}$, so $\mathcal{S}_{115} < B/2$ for small z. The initial displacement is $O(z^{14\mu_0-2}) \le \rho_z/4$. This verifies the finite initial conditions.

At every step set $t_{n+1} = t_n^{11/9}$. Taking t_0 small gives $\sum_n t_n < 1/4$. Equation (16) keeps the entire construction inside the permitted C^2 ball. Hold k fixed and use $\delta' = \tau^k$. For any fixed order m, even one above $r(k)$, (12) gives

$$x_{n+1} \le B/4 + c_n(1 + x_n), \qquad c_n = C_m(z,k)t_n^{2/9} \longrightarrow 0. \tag{17}$$

All x_n are finite since every finite-stage map is smooth. Once $c_n \le 1/4$, this first bounds the sequence; then its limsup is at most $B/4$. In particular, for $m = r(k+1)$ its bound eventually falls below B. Wait also for the next block's threshold and $C_{r(k+1)}(z,k)t^{2/9}(1+B) \le B/2$. Switching once to $\delta' = \tau^{k+1}$ leaves the new higher norm below $3B/4$. Thus each block has finite length, and k tends to infinity.

For any fixed m, eventually $\lfloor k/2 \rfloor \ge m$, so (16) is summable in C^m. The limit is smooth. The metric defects $\delta^2 H$ tend to zero in C^0; continuity of γ in C^1 proves exactness. Initial displacement plus the summable C^2 tail is at most $\rho_z/2$. Arbitrarily large higher initial derivatives affect the finite waiting times, not the stated uniform input assumptions. □

Corollary 4.3 (Conservative fixed-scale option)**.** *Keeping the source's original $\beta = 6/5$, $\lambda = 11/10$ and $k \ge 10$, one may take $r(k) = q(k) = 14k + 26$, hence $R = 166$, instead of $40(k+1)$ and $R = 440$. Its six exponents are*

$$1/5, \quad (3k-18)/5, \quad k+1/5, \quad (12k+14)/5, \quad 1/5, \quad (7k+14)/5.$$

The same initialization and block proof applies with $N > 20\mu_0 + 1$.

This option makes no change to the source's scale ratios or starting block. The stronger theorem explicitly rechecks them; neither result asserts that arbitrary finitely differentiable initial data become smooth, nor that C^{117} is an optimal geometric regularity threshold.

Proposition 4.4 (Optimality within this scale family)**.** *Among schemes using one common integer order $r = q$, a starting integer $k \ge 1$, $1 < \lambda < \beta$, and demanding all six exponents in* (11) *be at least $\beta - 1$, the smallest possible initial order is* 115*. This is an optimization of this sufficient error scheme, not a necessity theorem for exact isometric correction.*

Proof. The cubic row forces $\beta \le (3k+1)/(2k+4)$, so $k > 3$. The first and fifth rows force

$$r \min\{\beta - \lambda, \lambda - 1\} \ge (2\beta - 1)k + 3\beta - 1.$$

Since the minimum is at most $(\beta - 1)/2$,

$$r \ge 4k + 6 + \frac{2(k+2)}{\beta - 1} \ge 8k + 34 + \frac{100}{k-3}.$$

For integer $k = 4, 5, 6, 7$ the last values are $166, 124, 346/3, 115$; for $k \ge 7$ successive differences are $8 - 100/[(k-3)(k-2)] > 0$. Thus $r \ge 115$. The choice (14) at $k = 7$ attains all requirements, proving the claim. □

4.2 Observable Waiting and Tail Budgets

The recurrence also gives a conditional finite accounting formula. Write $X_n = 1 + x_n$, $D = B/4 + 1$, $c_n = Ct_n^{\beta-1}$ inside a fixed block. Iteration of (17) yields

$$X_N \le D \sum_{j=0}^{N-1} \prod_{i=j+1}^{N-1} c_i + X_0 \prod_{i=0}^{N-1} c_i, \qquad \prod_{i=j}^{N-1} c_i = C^{N-j} \frac{t_N}{t_j}. \tag{18}$$

The second identity follows from the geometric sum of the exponents $\beta^i(\beta - 1)$. It makes explicit how a large high-order input is forgotten after a sufficiently long block, if C and that input norm

are actually bounded. No effective waiting time is supplied by unspecified source constants alone. Once every later block controls C^m, the tail is bounded by

$$\sum_{j=0}^{\infty} \|U_{N+j}\|_{C^m} \le \rho_z \frac{t_N}{1 - t_N^{\beta-1}}. \tag{19}$$

Indeed $\beta^j \ge 1 + j(\beta - 1)$. Certifying a finite list of increments without the future block hypotheses does not certify this infinite tail.

5 Primitive Geometry and Later Boundaries

The preceding refinements do not dispose of the source's hardest geometric conditions. This section records the actual interfaces to avoid substituting a scalar convergence argument for them.

5.1 Velocity, Cutoff and Periodic Representation

In a primitive disk choose coordinates (x, y) with a good phase x and $\mathrm{Hess}_{\gamma(G)}\, x(\partial_y, \partial_y) > 0$. Let $Y = G_y$, $C = G_{yy}$ and $P = \mathrm{span}(Y, C)^{\perp}$. The Hessian condition gives $\Gamma^x_{yy} < 0$, making the tangential projections of Y, C independent; the normal plane projects isomorphically onto P. For $v = \mathrm{proj}_P G_x$ and a nonnegative amplitude a flat on the primitive boundary, the source constructs a smooth loop $V \in P$ with

$$|V|^2 = |v|^2 + a^2 =: R^2 > 0, \qquad \mathrm{av}\, V = v.$$

Then $X = G_x + V - v$ satisfies, exactly,

$$|X|^2 = |G_x|^2 + a^2, \qquad X \cdot Y = G_x \cdot G_y. \tag{20}$$

In the boundary collar the loop is a small arc with amplitude proportional to $a/|v|$, with a smooth even correction; it becomes constant smoothly where $a = 0$. In the interior a positive reparametrization density realizes the mean v. Its existence uses convex hull interior, smooth positive densities and a finite constraint inverse, not arbitrary replacement of the physical loop by a discrete quadrature law.

If $\mathcal{I}$ denotes the mean-zero periodic primitive, the leading corrector is $U_1 = \mathcal{I}(V - v)$. It satisfies $Y \cdot U_1 = C \cdot U_1 = 0$, hence $Y \cdot \partial_y U_1 = 0$. This cancellation removes the first problematic yy metric term. At later orders the tangential part $T \in \mathrm{span}(Y, C)$ is determined by a two-by-two Gram solve,

$$Y \cdot T = \mathcal{I}h, \qquad C \cdot T = \partial_y \mathcal{I}h - j.$$

The remaining equation for $U = T + \mathcal{I}W$ is exactly the mean-zero operator A of Section 3, with remainder $r = e - \Pi(X \cdot T_\theta)$. Thus its least-energy interpretation applies to the actual auxiliary equation, without changing its normalizations.

For any fixed integer L, the source obtains a finite expansion

$$f_z[G] = G + \sum_{j=1}^{L} z^j U_j(G; x/z), \qquad \gamma(f_z[G]) = a^2 dx^2 + P_z(G) + \sum_{j=L}^{2L} z^j R_j.$$

Here U_j uses jets through order $2j$, and the coefficients of P_z through order $2j + 2$. These are finite identities with controlled remainders, not a convergent infinite power series. A finite slow-map sequence uses (2) for P_z, with its exponent p independent of the finite accuracy. Orders and small powers are chosen sequentially so that the resulting f_z meets a finite exact-correction hypothesis. Theorem 4.2 reduces that hypothesis; it does not remove the need to produce the required approximate family or verify its geometric bounds.

5.2 Both Ordered Crossing Conditions

For a later boundary tangent Y_i, set $A_i = B(Y_i, Y_i)$. The source retains $A_i \neq 0$ and $A_i/|A_i| \neq -n$ for a preferred unit normal n. At pairwise boundary intersections it requires $\langle A_i, n\rangle > 0$ and, in both orders,

$$Q_{i\ell} = \left\langle B(Y_i, Y_\ell), \frac{A_i}{|A_i|} \right\rangle^2 + K|Y_i \wedge Y_\ell|^2_\gamma > 0. \tag{21}$$

The equality derived from Gauss is $Q_{i\ell} = \langle A_i, A_\ell\rangle - |\operatorname{proj}_{A_i^\perp} B(Y_i, Y_\ell)|^2$. It explains why merely preserving $B \neq 0$ or one preferred-normal pairing is insufficient.

In the adapted normal frame write

$$B_{yy} = (S, 0), \quad B_{xy} = (D, N), \quad B_{xx} = ((\kappa + D^2 + N^2)/S, L), \quad \kappa = K \det\gamma, \quad S > 0. \tag{22}$$

Gauss proves the first component of B_{xx}, without controlling L. For $Y_i = b_i\partial_x + c_i\partial_y$,

$$S\,\langle A_i, n_*\rangle = (Db_i + Sc_i)^2 + (\kappa + N^2)b_i^2.$$

If $b_i \neq 0$ and $t_i = c_i/b_i$, the scaled vectors at a crossing are

$$\begin{aligned} P_i &= ((N^2 + \kappa + (D + St_i)^2)/S,\ L + 2Nt_i), \\ Z_{i\ell} &= ((N^2 + \kappa + (D + St_i)(D + St_\ell))/S,\ L + N(t_i + t_\ell)), \\ Z_{i\ell} - P_i &= (t_\ell - t_i)(D + St_i, N). \end{aligned}$$

The last identity is independent of the potentially large L. The source uses it in the projections underlying (21), together with turn regions where $|N|$ is large and nonturn regions where $|L|$ grows like z^{-1}. The turn parameter is fixed before taking z small. Its support construction treats the finitely many tangencies separately. This sequence of choices, and the two ordered inequalities, remain geometric obligations; the periodic energy formula does not prove them.

Finally the preferred normal is modified near crossings and interpolated using nonantipodality, then compared with the projected old normal outside the primitive disk. The exact correction preserves these strict conditions by C^2 proximity after their margins have been fixed. This is not an exact boundary-value inverse theorem.

6 Global Assembly, Limits and Useful Extensions

The source begins with an ordinary immersion $I : M \to S^3$, importing the underlying immersion theorem. On a finite crossing set it uses disjoint quadratic bumps to kill B^{S^3} there, with uniform C^2 bounds, then normalizes back to S^3. For $F_0 = \lambda I$ the radial normal $n = -I$ is global, even when M is nonorientable, and

$$B_{F_0} = \lambda B_I^{S^3} + \lambda^{-1}\gamma(F_0)n.$$

At the prepared crossings the radial contribution makes the required pairings and $Q_{i\ell}$ positive. This is one global normal section, not a claim that a global oriented normal frame has been constructed.

A finite number of exact metric cycles is selected before the actual boundary crossing set. The coefficient operator for each cycle is close to the identity and invertible; its amplitudes give exactly the intended h/N_c contribution. Partial cycles stay close to the reference path when N_c is large. Generic radius choices give pairwise transverse circles, no triple crossings, and the finite tangency/direction conditions. Only then is the initial immersion prepared at the resulting crossing set. Uniform C^2 control, not uniform all-order control over those preparations, avoids a circular dependence. The finite primitive induction still requires every geometric conclusion summarized in Section 5.

6.1 Advantages, Costs and Nonclaims

The order reduction lowers the number of *uniform family* derivatives needed to start exact correction, and the common-order optimality calculation shows precisely what is being optimized. It does not measure runtime, reduce a geometric target dimension, prove a finite-regularity version of the global theorem, or bound the waiting time when higher norms are unknown. The constant N and thresholds depend on source polynomial estimates; no numeric universal N has been computed here. The least-energy formula gives a physically normalized correction cost and detects nearly singular loops. Its cost can diverge as $\mu \downarrow 0$, and its favorable L^2 norm does not automatically improve the spatial C^m bounds used by the geometry.

Two elementary warnings are relevant. On a periodic coordinate, $u_j(x) = j^{-2}\sin(jx)$ tends to zero uniformly while u_j'' does not; small undifferentiated residuals do not preserve curvature or boundary inequalities. Also, finite point evaluations can miss any smooth bump supported between sample points. Neither a finite moment fit nor a sampled metric equality is an all-domain residual enclosure. Original Ram recovery and stability results require the missing remainder and inverse-domain hypotheses explicitly.

Feasible next steps are to derive whole-parameter enclosures for Σ and its spatial derivatives for the actual constructed loop; to optimize separate moment and accuracy orders instead of imposing $r = q$; and to make the polynomial phase constants explicit enough to bound finite waiting times. Each requires additional analysis. A support-preserving exact correction would require a different theorem, not just smaller residuals. No claim of such a theorem is made here.

7 Read Scope and Executable Checks

All thirteen assigned public text files were displayed in bounded, untruncated chunks, totaling 3168 lines. This includes all mathematical modules, the figure source, main file, bibliography and README. Original private reading comprised the complete 295-line stability chapter and 569 lines of the 1885-line jet chapter: 1–170, 495–606, 730–847, 883–943, and 1195–1302. The complement in the jet chapter is explicitly unread. These are fresh reads, not parent or earlier report counts. Pre-read hashes were persisted before display; final hashes and exact interval complements are recorded in the owned read ledger. Captured publisher-tree byte identity is provenance, not proof validity or a fresh live-branch assertion. No whole private corpus audit is claimed.

The owned executable suite contains 86 exact rational/symbolic cases and 487 assertions, with zero numerical cases and zero parent or foreign cases. It checks the six exponents and admissibility margins, the fixed-scale alternative, the architecture lower bound, physical weighted periodic solutions and nullspace energies, the singular two-point negative control, the true-map quadratic cancellation, signed moment systems and block products. The fixtures do not establish the geometric phase conditions, uniform spatial derivatives, infinite convergence or global assembly. Written proofs, with their stated source dependencies, carry those local deductions. No source mathematical correction was found in the portions rederived here; the refinements concern sufficient input and cost bounds.

References

[1] OpenAI, *Smooth isometric immersions of closed surfaces into Euclidean four-space*, September 23, 2026, assigned cached manuscript, Publisher manuscript directory. Captured source, all thirteen text files read; global conclusion not independently certified by this report.

[2] M. Lewicka, *Full flexibility of isometric immersions of metrics with low Hölder regularity in Poznyak theorem's dimension*, arXiv:2511.16305v3, August 14, 2026, `https://arxiv.org/abs/2511.16305v3`. Fresh metadata, Theorem 1.1 and Appendix Theorem 9.1/proof sketch read; not the complete external proof.

[3] J. Moser, *A rapidly convergent iteration method and non-linear partial differential equations. I*, Ann. Scuola Norm. Sup. Pisa **20** (1966), 265–315, `https://numdam.org/item/ASNSP_1966_3_20_2_265_0/`. Classical context; no complete external proof audit asserted.

[4] M. Subbarao, original confidential Ram nonlinear integral-equation/PDE manuscript, TR1, `stability.tex`, lines 1–295. Local original proof source for bounded-isomorphism radii, perturbation, residual and transfer hypotheses; not uploaded.

[5] M. Subbarao, original confidential revised Ram nonlinear integral-equation/PDE manuscript, TR1, `sec-03-jets.tex`; literal intervals listed in Section 7. Original Taylor, Leibniz, differentiated-remainder and finite-recovery passages; unread parts and other corpus files not claimed as audited.